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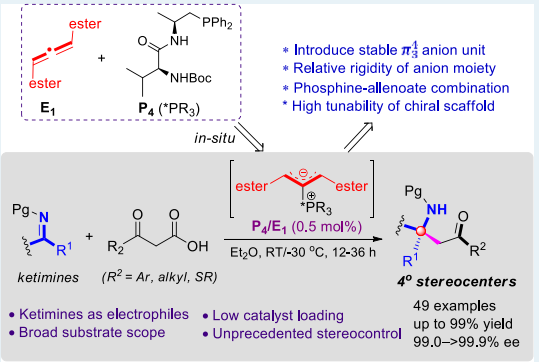
Highly Enantioselective Construction of Fully Substituted Stereocenters Enabled by *In Situ* Phosphonium-Containing Organocatalysis

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ABSTRACT: We present, herein, an important and practical class of bifunctional *in situ* phosphonium-containing organocatalysis and its initial application in highly efficient and enantioselective construction of N-containing fully substituted stereocenters through an asymmetric decarboxylative transformation between cyclic ketimines and β-keto acids. With this catalytic protocol, a variety of optically active cyclic tertiary amine derivatives were readily synthesized in high yields with excellent enantioselectivities (99.0 to >99.9% ee) under very low catalyst loading (0.5 mol %). The success of gram-level preparation and synthetic transformations proves the potential of this catalytic strategy for practical applications. Mechanistic investigation, including control experiments, *in situ* NMR analysis, and ESI-HRMS characterization of intermediates, provides insights into the mechanism.



KEYWORDS: *fully-substituted stereocenter, decarboxylative addition, phosphine-allenoate organocatalysis, Brønsted base, reaction mechanism*

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INTRODUCTION

Over the past decades, nucleophilic organophosphine catalysis has emerged as a powerful approach for the creation of structurally diverse and synthetically valuable chiral molecules in organic chemistry.[1,2] In general, phosphine-directed catalytic events typically started with nucleophilic addition of a tertiary phosphine catalyst to an electron-deficient reactant such as alkene, allene, and alkyne, forming the key zwitterionic intermediate which can further react with a variety of electrophilic reactants to furnish different enantioselective chemical transformations.[2] Notably, such zwitterion intermediates generated upon phosphine addition are basic in nature[3−5] and can thus be utilized as a new class of organic base catalysts in asymmetric synthesis. Accordingly, phosphine-containing Brønsted base catalysis has recently been developed. As a particularly successful motif, Zhao and coworkers introduced their pioneering investigations on phosphine-acrylate dual-reagent catalysis using Mannich-type[6a−f] and Strecker[6g] reactions for defining the state of this art, in which the zwitterionic intermediates of type **A** *in situ* generated from amino acid-based phosphine and acrylate were rightfully utilized as organic base catalysts (Figure 1a). However, despite such impressive progress has been made, important challenges remain. Among them are (1) the evaluation of the previously unexplored anion moiety of catalysts, (2) the development of further high efficient catalysts, and (3) the extension of the reaction scope.

In this study, we attempted to develop an elegant and complementary phosphine-containing organic base catalysis toward meeting these challenges. Indeed, the intrinsic structural weakness of phosphine-acrylate catalyst **A** is that this type of zwitterions inevitably bears a formally rotatable sp^3 C−C bond between the phosphonium cation and carbon anion, which makes the chiral scaffold much more flexible than the precatalyst phosphine. To address this issue, a structurally rigid anion scaffold needs to be introduced during the catalyst design.[7] We reasonably speculate that allenoate is probably a promising alternative used as anion sources in combination with amino acid-based bifunctional phosphine as a precatalyst because of the following structurally natural advantages of zwitterions of type **B** (Figure 1b). First, this *in situ* Brønsted base catalyst (**B**) contains a particular π_3^4 anion unit, which makes that the skeleton of phosphine-allenoate catalyst **B** is relatively more rigid than that of phosphine-acrylate catalyst **A**

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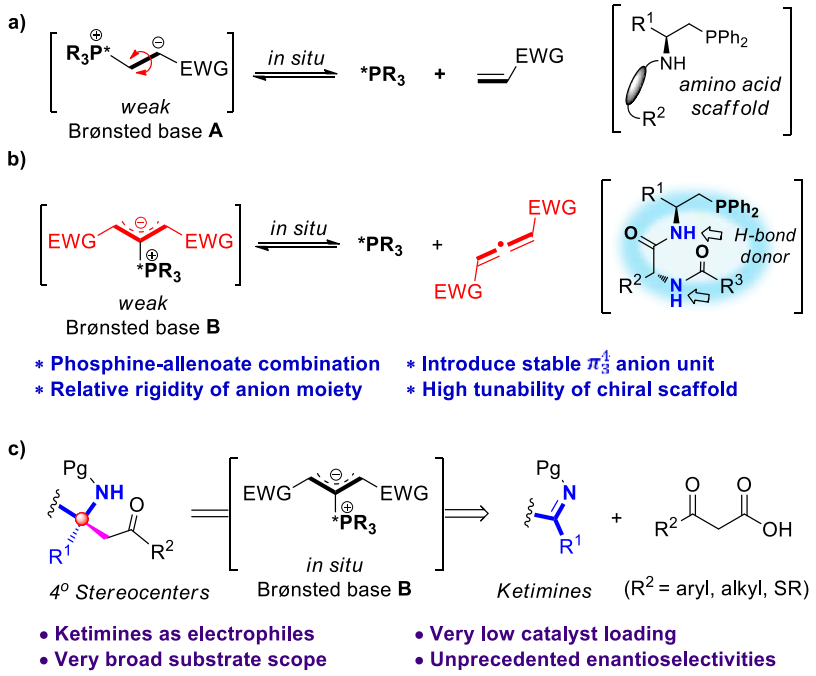


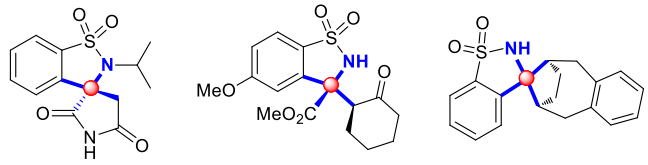
Figure 1. *In situ* phosphonium-containing organocatalysis and its application to asymmetric construction of N-containing fully substituted stereocenters. (a) Previous *in situ* phosphine-acrylate organocatalysts. (b) This new *in situ* phosphine-allenoate organocatalysts. (c) Asymmetric construction of N-containing fully substituted stereocenters by *in situ* phosphine-allenoate organocatalysis.

and can thus be favorable to stereoinduction in organic transformations. Second, the double electron-withdrawing ester groups located in the anion moiety further enhance the stability and reactivity of the negatively charged active site of the catalyst. Additionally, the cation moiety of amino acid-derived phosphines[1] not only possesses remarkably high tunability but also contains properly installed hydrogen-bonding donors, which would possibly lead to an enhanced stereochemical communication between the catalyst and substrate. With these views in mind, we have identified the allenoate as a particularly promising alternative for new phosphine-containing organic base catalysis. In the first example, we describe that a sufficiently active and unprecedentedly stereoselective phosphine-allenoate catalyst enabled asymmetric construction of N-substituted quaternary stereocenters via a decarboxylative addition of cyclic ketimines and β-keto acids.

This decarboxylative Mannich-type reaction, which mimics the reaction pathway of natural biosynthesis, was shown to be one of the most straightforward and efficient approaches for producing functionalized chiral amines.[8] The reaction was typically catalyzed by Lewis acid and chiral amine catalysts, and important advances have been achieved so far.[9,10] Nevertheless, the asymmetric versions on decarboxylative addition of sterically hindered ketimines particularly as cyclic *N*-sulfonyl ketimines are very rare[9d−g,10a] despite the fact that such process provides a powerful tool for preparing biologically valuable but synthetically challenging N-containing fully substituted stereocenters,[11] which are also significant building blocks in numerous bioactive compounds (Scheme 1).[12] Furthermore, to the best of our knowledge, there is no report on an asymmetric decarboxylative addition process in application of phosphine-containing organic base catalysis.

On account of our continuous interest on developing novel phosphonium-involving organocatalysis,[13] we were fascinated

Scheme 1. Selected N-Substituted Quaternary Stereocenter-Containing Bioactive Cyclic Compounds



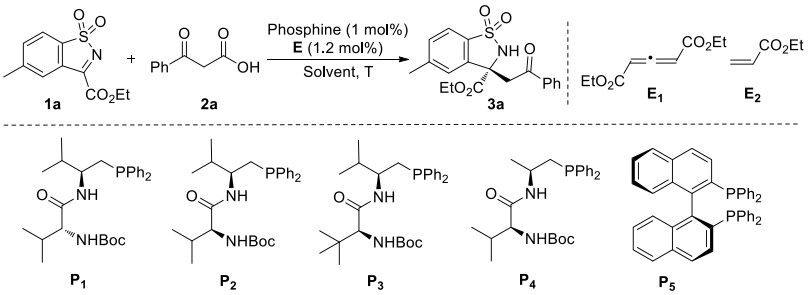
with the possibility that the aforementioned *in situ* bifunctional Brønsted base catalyst of phosphine-allenoate (**B**) might provide sufficient activation for the challenging ketimine substrates and thus realize a highly stereoselective decarboxylative addition using broad β-keto acids as reaction partners. Herein, we present a new class of dipeptide-based phosphine-allenoate catalyst and its first application example on catalytic asymmetric decarboxylative addition reaction between cyclic ketimines and β-keto acids (Figure 1c), thus leading to facile construction of N-containing fully substituted stereocenter systems.

RESULTS AND DISCUSSION

Asymmetric Decarboxylative Addition of Cyclic Ketimines with Aryl Substituted β-Keto Acids. Initially, we selected cyclic *N*-sulfonyl α-ketimino ester **1a** and β-keto acid **2a** as model substrates for optimizing reaction conditions (Table 1). The reaction was first performed in the presence of a L-Val-D-Val-based bifunctional phosphine catalyst $\mathbf{P_1}$ and allenoate $\mathbf{E_1}$ under ambient atmosphere with dichloromethane as a solvent, and the desired additive product **3a** was obtained in excellent yield (95%) but with disappointing enantioselectivity (3% ee) (entry 1). We were not discouraged by this result because it still verified that the catalytic ability of this

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Table 1. Optimization for Asymmetric Decarboxylative Addition of Ketamine 1a and Phenyl β-Keto Acid 2a[a]



entry	P/E	solvent	T (°C)	t (h)	yield (%)[b]	ee[c]
1	P_1/E_1	CH_2Cl_2	rt	8	95	3
2	P_2/E_1	CH_2Cl_2	rt	8	95	93
3	P_3/E_1	CH_2Cl_2	rt	8	94	95
4	P_4/E_1	CH_2Cl_2	rt	8	95	99
5	P_5/E_1	CH_2Cl_2	rt	8	94	1
6	P_4/E_1	$CHCl_3$	rt	8	95	85
7	P_4/E_1	toluene	rt	12	80	59
8	P_4/E_1	hexane	rt	20	94	57
9	P_4/E_1	Et_2O	rt	9	98	>99.9
10	P_4/E_2	Et_2O	rt	10	93	29
11[d]	P_4/E_1	Et_2O	rt	12	98	>99.9
12	P_4	Et_2O	rt	12	95	0
13	none	Et_2O	rt	55	68	0

[a]Reactions were performed with **1a** (0.10 mmol), **2a** (0.12 mmol), and phosphine/**E** (1 mol %, 1/1.2 mol ration) in the solvent (1.0 mL) at room temperature. [b]Isolated yield. [c]The ee values were determined by chiral HPLC analysis. [d]Only 0.5 mol % phosphine catalyst was used.

dual system is assured. Later, we examined the combination of L-Val-L-Val- or L-Val-L-*tert*-Leu-based phosphine catalysts (P_2 or P_3) and allenoate E_1. To our delight, both excellent yield and ee value were achieved (entry 2). Further minor modification of phosphine catalysts could make the enantioselectivity to be slightly improved, and P_4 was finally proved to be the best chiral phosphine (entries 3 and 4). Of note, the monofunctional phosphine (*S*)-BINAP (P_5) was screened to give the desired product in an almost racemic form, which implied that the hydrogen-bonding interaction is crucial for stereocontrol (entry 5). Later, the solvent effect was evaluated under the optimized catalyst of P_4/E_1 combination (entries 6−9). It was found that diethyl ether showed the highest performance toward both reactivity and enantioselectivity (entry 9). By comparison, when acrylate E_2 was used as a co-catalyst alternative for chiral phosphine P_4, the reaction worked and afforded the desired product **3a** with low ee (entry 10, see Table S6 in Supporting Information for details). This result strongly supported our hypothesis that flexibility of the phosphine-acrylate catalyst would dramatically decrease the enantioselectivity. Notably, reducing the catalyst loading to 0.5 mol % could also afford excellent reaction results (98% yield and >99.9% ee) (entry 11). Moreover, the racemic product was formed in the absence of either allenoate E_1 or both P_4 and E_1, which suggested that the *in situ* generated zwitterion was the catalytically active species for promoting this enantioselective decarboxylative addition (entries 12 and 13).

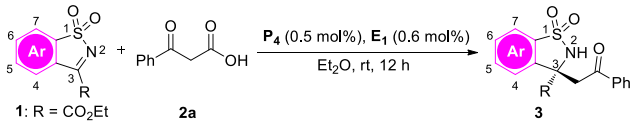
With the optimized reaction conditions identified (entry 11, Table 1), a large range of cyclic *N*-sulfonyl ketimines bearing different substituent groups were investigated. In all cases, the reaction proceeded smoothly and afforded the desired product in good to excellent yields (83−99%) with unprecedented enantioselectivities (Table 2). It is worth noting that all desired products were obtained with very excellent enantioselectivities (99.0 to >99.9% ee) irrespective of the electronic nature and positions of substituents. The reactivity of the title reaction is generally high under phosphine-allenoate catalysis. In the case of substrates bearing electron-withdrawing groups such as either F or CF_3 at the 5-position, the yields of desired products (**3g** and **3h**) were slightly lower than others (entries 7 and 8).

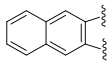
Subsequently, the catalytic system was also examined for the decarboxylative addition reactions of various aryl-substituted β-keto acids (Table 3). To our surprise, excellent yields (91−98%) with unprecedented stereoselectivities (99.0 to >99.9% ee) were achieved for all entries. The phosphine-allenoate (P_4/E_1) combination seems to be an "ideal" catalytic system for this enantioselective chemical process, in which the electronic and steric effects of substituents on the aromatic ring have little influence on both catalytic reactivity and enantioselectivity under this dual catalysis. In addition, the malonic acid half thioester **2t** was also a suitable substrate to give the corresponding adduct **4t** in 98% yield with >99% ee under the standard reaction conditions using dichloromethane as the solvent alternatively.

Asymmetric Decarboxylative Addition between Cyclic Ketimines and Alkyl-Substituted β-Keto Acids. Compared with aryl-substituted β-keto acids, alkyl-substituted β-keto acids are less studied in such decarboxylative addition reactions owing to their intrinsic instability toward most catalytic environment. The aliphatic β-keto acids have a greater tendency to undergo the decarboxylative process very rapidly before and after the desired addition process, which may cause low efficiency and stereoselectivity.[11c] Indeed, while the decarboxylative addition of 3-oxopentanoic acid **5a** with cyclic ketimine **1a** was performed under the above-mentioned standard conditions, it was found that the yield maintained with a longer reaction time (24 h), but the enantioselectivity dropped to 38% ee (Table 4, entry 4). To improve the

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Table 2. Substrate Scope of Ketimines Reacted with 2a[a]

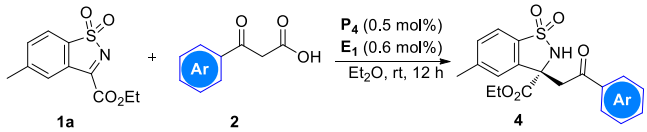


entry	Ar	3	yield (%)[b]	ee[c]
1	5-Me-C_6H_3	**3a**	98	>99.9
2	C_6H_4	**3b**	98	>99.9
3	5-OMe-C_6H_3	**3c**	98	>99
4[d]	5-^{t}Bu-C_6H_3	**3d**	96	99
5	5-Ph-C_6H_3	**3e**	94	>99.9
6	5-Cl-C_6H_3	**3f**	99	>99
7	5-F-C_6H_3	**3g**	86	>99
8	5-CF_3-C_6H_3	**3h**	83	99
9	5-OCF_3-C_6H_3	**3i**	96	>99
10	6-Me-C_6H_3	**3j**	99	>99
11	6-Cl-C_6H_3	**3k**	97	>99.9
12	7-Cl-C_6H_3	**3l**	98	>99
13	7-F-C_6H_3	**3m**	96	>99
14		**3n**	98	>99.9
15		**3o**	98	>99.9

[a]Reactions were performed with **1** (0.10 mmol), **2a** (0.12 mmol), and $\mathbf{P_4}/\mathbf{E_1}$ (0.5 mol %, 1/1.2 mol ratio) in Et_2O (1.0 mL) at room temperature for 12 h. [b]Isolated yield. [c]The ee values were determined by chiral HPLC analysis. [d]The CH_2Cl_2 was used as solvent.

enantioselectivity of this transformation, we further examined other phosphines by combining allenoate $\mathbf{E_1}$ as the co-catalyst and found that the phosphine $\mathbf{P_4}$ was also the choice of the co-catalyst (entries 1−5). Later, solvent screening was performed (entries 6−12), and the Et_2O was found to be the solvent of choice on affording the desired product in 95% yield and 85% ee at room temperature. In addition, increasing catalyst loading and decreasing the reaction temperature were proved to be effective approaches for furnishing the corresponding product in 92% yield and >99% ee (entry 14). Encouraged by these results, we set our goal to extend this novel strategy to asymmetric decarboxylative addition reactions of aliphatic β-keto acids **5** together with cyclic ketamines **1**. As shown in Table 5, the corresponding addition reaction was performed in the presence of 1 mol % catalyst ($\mathbf{P_4}/\mathbf{E_1}$) under −30 °C. As expected, all entries provided satisfying results, affording desired adducts in excellent isolated yields (91−96%) and enantioselectivities (99.0 to >99.9% ee).

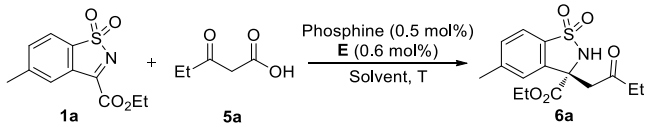
Table 3. Substrate Scope of Aryl-Substituted β-Keto Acids Reacted with Ketimine 1a[a]



entry	Ar	4	yield (%)[b]	ee[c]
1	2-Me−C_6H_4	**4a**	97	>99.9
2	2-MeO−C_6H_4	**4b**	96	>99
3	2-F−C_6H_4	**4c**	94	>99
4	2-Cl−C_6H_4	**4d**	95	>99.9
5	3-Me−C_6H_4	**4e**	96	>99
6	3-MeO−C_6H_4	**4f**	96	>99.9
7	3-F−C_6H_4	**4g**	95	>99.9
8	3-Cl−C_6H_4	**4h**	96	>99.9
9	3-Br−C_6H_4	**4i**	95	99
10	4-Me−C_6H_4	**4j**	98	>99
11	4-MeO−C_6H_4	**4k**	97	>99.9
12	4-F−C_6H_4	**4l**	92	>99.9
13	4-Cl−C_6H_4	**4m**	93	>99.9
14	4-Br−C_6H_4	**4n**	93	>99.9
15	4-CF_3−C_6H_4	**4o**	91	>99.9
16	2,4-Me−C_6H_3	**4p**	96	99.9
17	1-naphthyl	**4q**	96	>99
18	2-naphthyl	**4r**	95	>99.9
19	2-thienyl	**4s**	97	>99
20[d]	−SC_6H_5	**4t**	98	>99

[a]Reactions were performed with **1a** (0.10 mmol), **2** (0.12 mmol), and $\mathbf{P_4}/\mathbf{E_1}$ (0.5 mol %, 1/1.2 mol ratio) in Et_2O (1.0 mL) at room temperature for 12 h. [b]Isolated yield. [c]The ee values were determined by chiral HPLC analysis. [d]The CH_2Cl_2 was used as the solvent.

Table 4. Optimization for Asymmetric Decarboxylative Addition of Ketimine 1a with Alkyl-Substituted β-Keto Acid 5a[a]

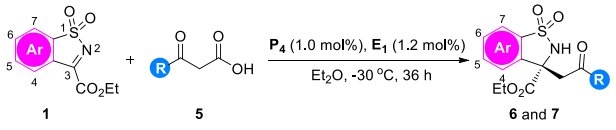


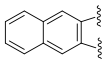
entry	$\mathbf{P}/\mathbf{E_1}$	solvent	t (h)	yield (%)[b]	ee[c]
1	$\mathbf{P_1}/\mathbf{E_1}$	CH_2Cl_2	24	93	9
2	$\mathbf{P_2}/\mathbf{E_1}$	CH_2Cl_2	24	94	33
3	$\mathbf{P_3}/\mathbf{E_1}$	CH_2Cl_2	24	96	23
4	$\mathbf{P_4}/\mathbf{E_1}$	CH_2Cl_2	24	96	38
5	$\mathbf{P_5}/\mathbf{E_1}$	CH_2Cl_2	24	93	0
6	$\mathbf{P_4}/\mathbf{E_1}$	$CHCl_3$	24	95	11
7	$\mathbf{P_4}/\mathbf{E_1}$	toluene	28	89	13
8	$\mathbf{P_4}/\mathbf{E_1}$	xylene	28	95	5
9	$\mathbf{P_4}/\mathbf{E_1}$	1,4-dioxane	28	94	67
10	$\mathbf{P_4}/\mathbf{E_1}$	hexane	28	91	81
11	$\mathbf{P_4}/\mathbf{E_1}$	EA	12	92	17
12	$\mathbf{P_4}/\mathbf{E_1}$	Et_2O	24	95	85
13[d]	$\mathbf{P_4}/\mathbf{E_1}$	Et_2O	28	95	90
14[d,e]	$\mathbf{P_4}/\mathbf{E_1}$	Et_2O	36	92	>99

[a]Reactions were performed with **1a** (0.10 mmol), **5a** (0.12 mmol), and $\mathbf{P_4}/\mathbf{E_1}$ (0.5 mol %, 1/1.2 mol ratio) in dry Et_2O (1.0 mL) at room temperature. [b]Isolated yield. [c]The ee values were determined by chiral HPLC analysis. [d]The $\mathbf{P_4}/\mathbf{E_1}$ (1 mol %, 1/1.2) was used. [e]The reaction was performed at −30 °C.

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Table 5. Asymmetric Decarboxylative Reaction of Ketimines with Alkyl-Substituted β-Keto Acids[a]

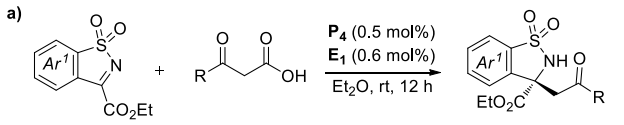


entry	Ar (1)	R (5)	6/7	yield (%)[b]	ee[c]
1	**1a**	Et (**5a**)	**6a**	92	>99
2	**1a**	Me	**6b**	92	>99.9
3	**1a**	*n*-Pr	**6c**	93	>99.9
4	**1a**	*i*-Pr	**6d**	93	>99.9
5	**1a**	*c*-Pr	**6e**	92	>99.9
6	**1a**	*n*-Bu	**6f**	94	>99.9
7	**1a**	*c*-hexyl	**6g**	93	>99
8	C_6H_4	**5a**	**7a**	94	>99
9	5-MeO-C_6H_3	**5a**	**7b**	95	>99
10	5-^{t}Bu-C_6H_3	**5a**	**7c**	96	>99.9
11	5-Cl-C_6H_3	**5a**	**7d**	92	99
12	7-Cl-C_6H_3	**5a**	**7e**	92	>99
13		**5a**	**7f**	92	>99
14		**5a**	**7g**	91	>99.9

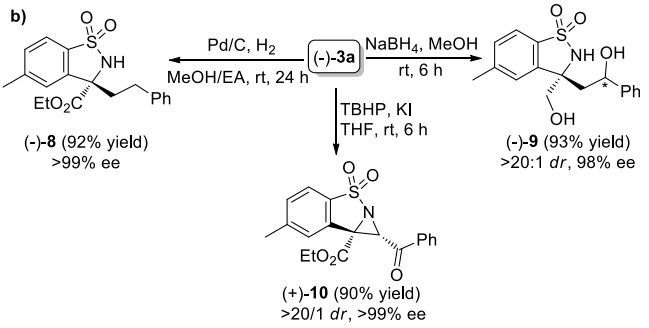
[a]Reactions were performed with **1** (0.10 mmol), **5** (0.12 mmol), and $\mathbf{P_4}/\mathbf{E_1}$ (1 mol %, 1/1.2) in Et_2O (1.0 mL) at −30 °C for 36 h. [b]Isolated yield. [c]The ee values were determined by chiral HPLC analysis.

Scaled-Up Synthesis and Transformations of the Product. To demonstrate the scalability and the practicality of the process, scaled-up preparation and synthetic transformations of the product were performed. As shown in Scheme 2a, scaled-up reactions of cyclic ketimine **1a** with different β-keto acids were performed, and the corresponding products were isolated in high yields without any loss of ee values. Furthermore, selective reductions of the title product **3a** using either Pd/C or $NaBH_4$ reagent could afford different chiral compounds (**8** and **9**) in high yields and excellent ee values, respectively. Also, treatment of **3a** with TBHP and KI led to simultaneous cyclization reactions for furnishing the aziridine compound ((+)-**10**), which are prevalent building blocks in many biologically active compounds.[14] In addition, the corresponding derivative (−)-**10** was also prepared (Schemes S1 and S2), and the absolute configuration of the (−)-**10** was assigned to be (1*R*, 7b*S*) by X-ray crystallographic analysis of its single crystal (Scheme 3).[15] Accordingly, the absolute configuration of the product (+)-**3a** was assigned to be *R*, and the absolute configuration of the product (−)-**3a** was thus assigned to be *S*. The configurations of other decarboxylative addition products were accordingly assigned by analogy.

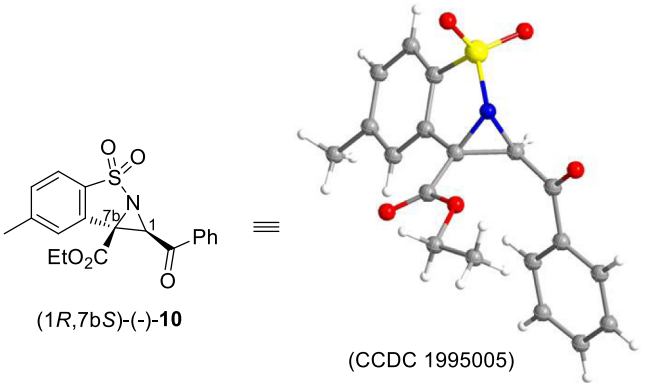
Scheme 2. Scaled-Up Reactions and Synthetic Transformations of the Decarboxylative Adduct; (a) Scaled-Up Reactions; and (b) Synthetic Manipulations of Product 3a



ketimine	β-keto acid	product	yield (%)	ee (%)
1a (3.0 mmol)	**2a**	(-)-**3a**	98	>99
1a (4.0 mmol)	**2k**	(-)-**4k**	95	>99.9
1a (4.0 mmol)	**5a**	(-)-**6a**	88	>99



Scheme 3. X-ray Crystal Structure of (−)-10[15]



Mechanistic Insights. The mechanistic reaction cycle of this decarboxylative addition is also believed to follow the general mechanism described in the literature,[10,11] and the detailed reaction pathway is shown in Figure 2. First, the nucleophilic addition of phosphine to the electron-deficient allenoate generates the zwitterion **B**, which can further deprotonate the incoming β-keto acid **2a** to form an ion-pair **I** that contains the phosphonium intermediate **C** and the anionic form of the nucleophile **D**. Then, addition of **D** to the ketimine **1a** leads to anionic **E** which may be stabilized by phosphonium **C** (ion-pair **II**). The intermediate **E** abstracts a proton from β-keto acid **2a** to generate the key intermediate In-**III** by forming the corresponding anionic **D** which will go into the reaction cycle again. At last, decarboxylation of the key intermediate In-**III** affords the desired product **3a**.

To gain a better understanding of this reaction cycle, *in situ* NMR and ESI/MS methods were used to study this decarboxylative addition reaction.[16] First, we conducted ^{31}P

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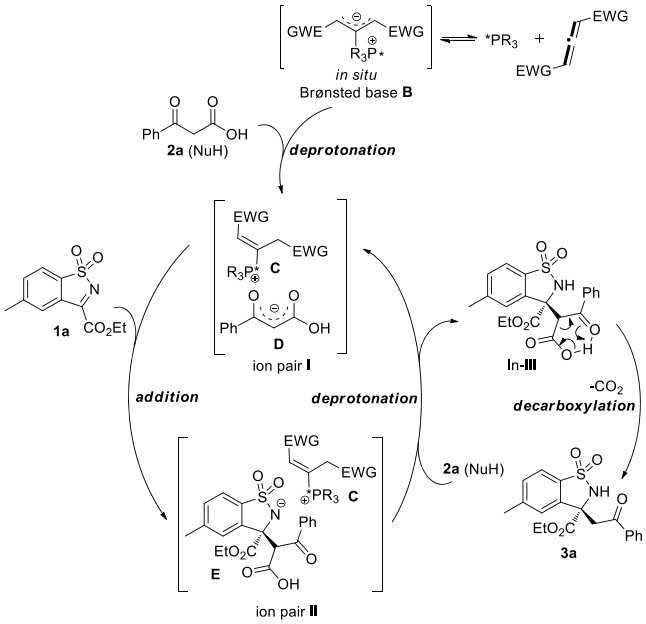


Figure 2. Proposed reaction cycle for this phosphine-allenoate catalyzed decarboxylative addition reaction.

NMR spectroscopic analysis of the reaction process in $CDCl_3$ (Figure 3a). The phosphine catalyst $\mathbf{P_4}$ alone showed a P-signal at −24.08 ppm (spectra **I**). When 1.2 equiv of the allenoate $\mathbf{E_1}$ was added to the phosphine $\mathbf{P_4}$, a new signal (at 27.12 ppm) appeared (spectra **II**), indicating that the Brønsted

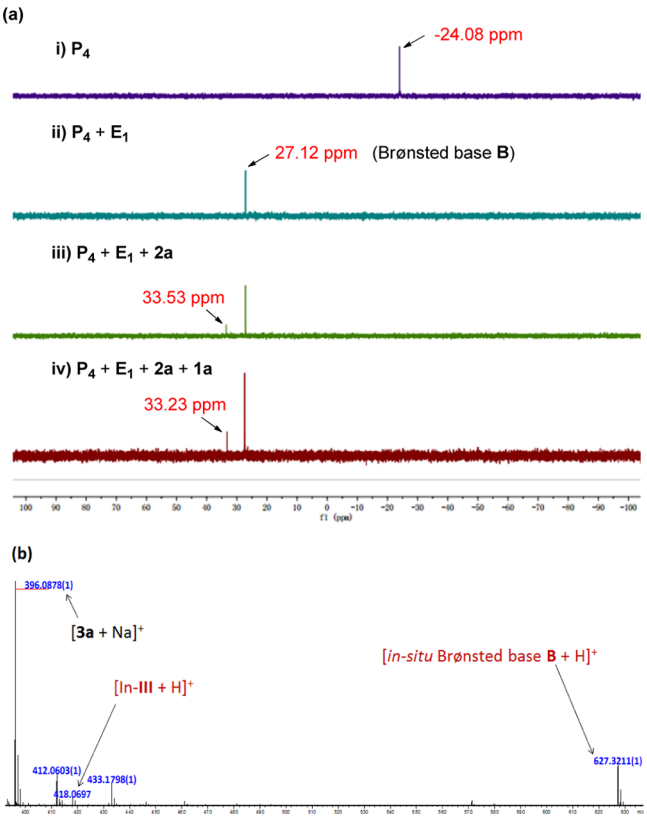
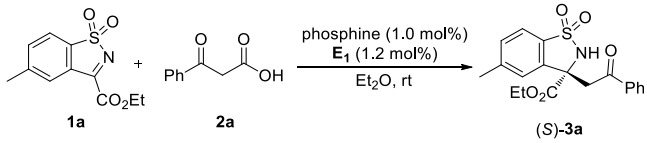


Figure 3. *In situ* ^{31}P NMR and ESI-HRMS studies on the reaction cycle. (a) ^{31}P NMR spectra. (b) ESI-HRMS spectrum (positive mode) of the reaction mixture.

base **B** was formed. When either *β*-keto acid **2a** or both *β*-keto acid **2a** and ketimine **1a** was added to the above-mentioned mixture of $\mathbf{P_4}/\mathbf{E_1}$, new peaks at 33.53 and 33.23 ppm were appeared, respectively, which might be tentatively assigned as a phosphonium-containing ion pair of type **I** or type **II** (spectrum **III** and **IV**). Notably, when either phosphine $\mathbf{P_4}$ or allenoate $\mathbf{E_1}$ was mixed with the solution of *β*-keto acid **2a** and cyclic ketimine **1a** alone, no appreciable P-signals can be observed. In addition, the high resolution mass spectroscopy (HRMS-ESI) analysis of the model reaction under optimal conditions was carried out (Figure 3b). Three signal peaks were detected at *m*/*z* 396.0878, 418.0697, and 627.3211, which corresponded to the decarboxylative product (**3a**), addition intermediate (In-**III**), and *in situ*-generated Brønsted base catalyst (**B**), respectively. These results offered solid support for our proposed weak Brønsted base catalytic mechanism and also illuminated that a stepwise process could be involved in the catalytic cycle (Figure 2), in which nucleophilic addition of the deprotonated ketoacid (**D**) to ketimine gave the key reaction intermediate (In-**III**), followed by the decarboxylation process to afford the desired product.

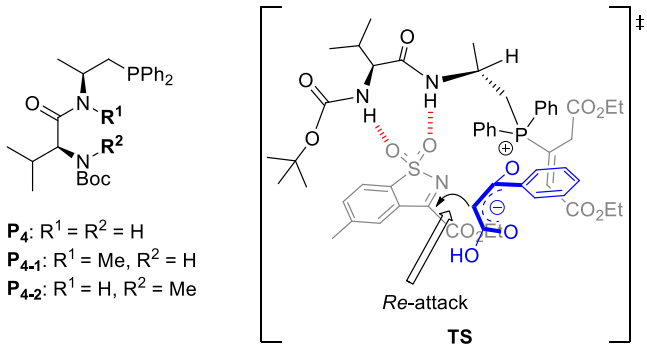
On the basis of the obtained results and our previous studies,[13] a plausible stereocontrol model is presented in Table 6. This sterically favored **TS** allows the nucleophilic attack of *β*-keto acids from the *Re* face of cyclic ketimines which predominantly delivers the *S* enantiomer. To provide experimental support, we prepared methylated catalysts $\mathbf{P_{4\text{-}1}}$ and $\mathbf{P_{4\text{-}2}}$ and tested their catalytic abilities for the model reaction. As shown in Table 6, when methylated phosphines were used as the co-catalyst, respectively, the reaction became slightly slower, and the enantioselectivities decreased dramatically. These experimental results clearly supported the

Table 6. Enantioselective Decarboxylative Addition of Ketimine 1a with *β*-Keto Acid 2a Promoted by Different Catalysts and the Proposed Stereochemical Model[a]



entry	1	phosphine	*t* (h)	yield (%)[b]	ee[c]
1	**1a**	$\mathbf{P_4}$	8	**3a**/98	>99.9
2	**1a**	$\mathbf{P_{4\text{-}1}}$	12	**3a**/97	17
3	**1a**	$\mathbf{P_{4\text{-}2}}$	12	**3a**/98	11

[a]The reactions were performed with **1a** (0.1 mmol), **2a** (0.12 mmol), and phosphine/$\mathbf{E_1}$ (1 mol %, 1/1.2 mol ratio) in dry Et_2O (1.0 mL) at room temperature. [b]Isolated yield. [c]The ee value was determined by HPLC analysis on chiral stationary phases.



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existence of hydrogen-bonding interactions between the dipeptide portion of the catalyst and imine substrates, which contributed significantly to the key transition state models. In addition, the activated ketoacid species associated closely with the phosphonium moiety of the catalyst through ionic interactions.

CONCLUSIONS

In summary, we have designed and developed a new type of dipeptide-based *in situ* phosphonium-containing organocatalyst and successfully achieved its application in the asymmetric construction of sterically hindered N-containing fully substituted stereocenters via an asymmetric decarboxylative Mannich process between β-keto acids and cyclic ketimines. In the presence of such a *in situ* bifunctional Brønsted base catalyst, the addition reactions proceeded smoothly for a broad variety of β-keto acids together with numerous cyclic ketimines under very low catalyst loading (0.5 mol %) and afforded series of functionalized tertiary amine-containing cyclic compounds in high yields with unprecedented enantioselectivities (99.0 to >99.9% ee). The practicality and the utility of this protocol were demonstrated by the scale-up synthetic reaction and facile elaborations. In addition, mechanistic investigations indicate that the reaction proceeds via a stepwise process of nucleophilic addition and subsequent decarboxylation. Further developments and applications of this novel *in situ* bifunctional phosphonium-containing organocatalyst to other challenging areas of organic synthesis are currently ongoing in our laboratory.

ASSOCIATED CONTENT

Supporting Information

The Supporting Information is available free of charge at https://pubs.acs.org/doi/10.1021/acscatal.0c01079.

Instrumentation and chemicals, optimization of reaction conditions, experimental procedure and characterization data for products, chiral HPLC conditions, scaled-up synthesis, synthetic transformation procedure, and NMR spectra (PDF)

Crystallographic data for compound (−)-**10** (CIF)

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Notes

The authors declare no competing financial interest.

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(15) See the Scheme S1 and S2 in Supporting Information for details, and CCDC 1995005 ((−)-**10a**) contain the supplementary crystallographic data for this paper. These data can be obtained free of charge from The Cambridge Crystallographic Data Centre.

(16) See Supporting Information for details.

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